

# Tidying Data

## tidyr

2026-07-30

# tidyr

Functions for tidying data.

What is tidy data?

“Tidy datasets are all alike, but every messy dataset is messy in its own way.” — Hadley Wickham



# Tidy Data

| country     | year | cases  | population |
|-------------|------|--------|------------|
| Afghanistan | 1999 | 745    | 19987071   |
| Afghanistan | 2000 | 2666   | 20593360   |
| Brazil      | 1999 | 37737  | 172006362  |
| Brazil      | 2000 | 80488  | 174504898  |
| China       | 1999 | 212258 | 1272915272 |
| China       | 2000 | 216766 | 1280425583 |

variables

| country     | year | cases  | population |
|-------------|------|--------|------------|
| Afghanistan | 1999 | 745    | 19987071   |
| Afghanistan | 2000 | 2666   | 20593360   |
| Brazil      | 1999 | 37737  | 172006362  |
| Brazil      | 2000 | 80488  | 174504898  |
| China       | 1999 | 212258 | 1272915272 |
| China       | 2000 | 216766 | 1280425583 |

observations

| country     | year | cases  | population |
|-------------|------|--------|------------|
| Afghanistan | 1999 | 745    | 19987071   |
| Afghanistan | 2000 | 2666   | 20593360   |
| Brazil      | 1999 | 37737  | 172006362  |
| Brazil      | 2000 | 80488  | 174504898  |
| China       | 1999 | 212258 | 1272915272 |
| China       | 2000 | 216766 | 1280425583 |

values

Each column is a single *variable*

Each row is a single *observation*

Each cell is a *value*

# pivot\_longer()

```
1 pivot_longer(<DATA>, <NAMES TO>, <VALUES TO>, <VARIABLES>)
```

# Lord of the Rings

```
1 lotr <- tribble(  
2           ~film,      ~race, ~female, ~male,  
3   "The Fellowship Of The Ring",  "Elf",  1229L,  971L,  
4   "The Fellowship Of The Ring", "Hobbit",   14L, 3644L,  
5   "The Fellowship Of The Ring",  "Man",    0L, 1995L,  
6           "The Two Towers",  "Elf",  331L,  513L,  
7           "The Two Towers", "Hobbit",   0L, 2463L,  
8           "The Two Towers",  "Man",  401L, 3589L,  
9   "The Return Of The King",  "Elf",  183L,  510L,  
10  "The Return Of The King", "Hobbit",   2L, 2673L,  
11  "The Return Of The King",  "Man",  268L, 2459L  
12 )
```

# Lord of the Rings

```
1 lotr
```

```
# A tibble: 9 × 4
```

|   | film                       | race   | female | male  |
|---|----------------------------|--------|--------|-------|
|   | <chr>                      | <chr>  | <int>  | <int> |
| 1 | The Fellowship Of The Ring | Elf    | 1229   | 971   |
| 2 | The Fellowship Of The Ring | Hobbit | 14     | 3644  |
| 3 | The Fellowship Of The Ring | Man    | 0      | 1995  |
| 4 | The Two Towers             | Elf    | 331    | 513   |
| 5 | The Two Towers             | Hobbit | 0      | 2463  |
| 6 | The Two Towers             | Man    | 401    | 3589  |
| 7 | The Return Of The King     | Elf    | 183    | 510   |
| 8 | The Return Of The King     | Hobbit | 2      | 2673  |
| 9 | The Return Of The King     | Man    | 268    | 2459  |



# new data alert!



## lotr

|   | film                       | race   | female | male |
|---|----------------------------|--------|--------|------|
| 1 | The Fellowship Of The Ring | Elf    | 1229   | 971  |
| 2 | The Fellowship Of The Ring | Hobbit | 14     | 3644 |
| 3 | The Fellowship Of The Ring | Man    | 0      | 1995 |
| 4 | The Two Towers             | Elf    | 331    | 513  |
| 5 | The Two Towers             | Hobbit | 0      | 2463 |
| 6 | The Two Towers             | Man    | 401    | 3589 |
| 7 | The Return Of The King     | Elf    | 183    | 510  |
| 8 | The Return Of The King     | Hobbit | 2      | 2673 |
| 9 | The Return Of The King     | Man    | 268    | 2459 |

Where does it come from?

exercises

source:

[github.com/jennybc/lotr-tidyr](https://github.com/jennybc/lotr-tidyr)

How can I use it?

Run the code at the top of  
exercises

```
View(lotr)
```



*this saves it in your  
global environment*

# pivot\_longer()

```
1 lotr |>
2   pivot_longer(
3     names_to = "sex",
4     values_to = "words",
5     cols = female:male
6   )
```



# pivot\_longer()

```
# A tibble: 18 × 4
```

|    | film                       | race   | sex    | words |
|----|----------------------------|--------|--------|-------|
|    | <chr>                      | <chr>  | <chr>  | <int> |
| 1  | The Fellowship Of The Ring | Elf    | female | 1229  |
| 2  | The Fellowship Of The Ring | Elf    | male   | 971   |
| 3  | The Fellowship Of The Ring | Hobbit | female | 14    |
| 4  | The Fellowship Of The Ring | Hobbit | male   | 3644  |
| 5  | The Fellowship Of The Ring | Man    | female | 0     |
| 6  | The Fellowship Of The Ring | Man    | male   | 1995  |
| 7  | The Two Towers             | Elf    | female | 331   |
| 8  | The Two Towers             | Elf    | male   | 513   |
| 9  | The Two Towers             | Hobbit | female | 0     |
| 10 | The Two Towers             | Hobbit | male   | 2463  |
| 11 | The Two Towers             | Man    | female | 101   |



# new data alert!



table2, table4a, who

|   | country     | iso2 | iso3 | year | new_sp_m014 |
|---|-------------|------|------|------|-------------|
| 1 | Afghanistan | AF   | AFG  | 1980 | NA          |
| 2 | Afghanistan | AF   | AFG  | 1981 | NA          |
| 3 | Afghanistan | AF   | AFG  | 1982 | NA          |
| 4 | Afghanistan | AF   | AFG  | 1983 | NA          |
| 5 | Afghanistan | AF   | AFG  | 1984 | NA          |
| 6 | Afghanistan | AF   | AFG  | 1985 | NA          |
| 7 | Afghanistan | AF   | AFG  | 1986 | NA          |
| 8 | Afghanistan | AF   | AFG  | 1987 | NA          |

|   | country     | 1999   | 2000   |
|---|-------------|--------|--------|
| 1 | Afghanistan | 745    | 2666   |
| 2 | Brazil      | 37737  | 80488  |
| 3 | China       | 212258 | 213766 |

|    | country     | year | type       | count      |
|----|-------------|------|------------|------------|
| 1  | Afghanistan | 1999 | cases      | 745        |
| 2  | Afghanistan | 1999 | population | 19987071   |
| 3  | Afghanistan | 2000 | cases      | 2666       |
| 4  | Afghanistan | 2000 | population | 20595360   |
| 5  | Brazil      | 1999 | cases      | 37737      |
| 6  | Brazil      | 1999 | population | 172006362  |
| 7  | Brazil      | 2000 | cases      | 80488      |
| 8  | Brazil      | 2000 | population | 174504898  |
| 9  | China       | 1999 | cases      | 212258     |
| 10 | China       | 1999 | population | 1272915272 |
| 11 | China       | 2000 | cases      | 213766     |
| 12 | China       | 2000 | population | 1280428583 |

Where does it come from?

The tidyr R package

How can I use it?

```
library(tidyr)
View(table2)
View(table4a)
View(who)
```



*they're  
invisible!*

## *Your Turn 1*

Use `pivot_longer()` to reorganize `table4a` into three columns: `country`, `year`, and `cases`.

# Your Turn 1

```
1 table4a |>
2   pivot_longer(
3     names_to = "year",
4     values_to = "cases",
5     cols = -country
6   )
```

# A tibble: 6 × 3

|   | country     | year  | cases  |
|---|-------------|-------|--------|
|   | <chr>       | <chr> | <dbl>  |
| 1 | Afghanistan | 1999  | 745    |
| 2 | Afghanistan | 2000  | 2666   |
| 3 | Brazil      | 1999  | 37737  |
| 4 | Brazil      | 2000  | 80488  |
| 5 | China       | 1999  | 212258 |
| 6 | China       | 2000  | 213766 |

# pivot\_wider()

```
1 pivot_wider(<DATA>, <NAMES FROM>, <VALUES FROM>)
```

wide

| id | x | y | z |
|----|---|---|---|
| 1  | a | c | e |
| 2  | b | d | f |

Animation by Mara Averick

# pivot\_wider()

```
1 lotr |>
2   pivot_longer(
3     names_to = "sex",
4     values_to = "words",
5     cols = female:male
6   ) |>
7   pivot_wider(
8     names_from = race,
9     values_from = words
10  )
```

```
# A tibble: 6 × 5
```

|   | film                       | sex    | Elf   | Hobbit | Man   |
|---|----------------------------|--------|-------|--------|-------|
|   | <chr>                      | <chr>  | <int> | <int>  | <int> |
| 1 | The Fellowship Of The Ring | female | 1229  | 14     | 0     |
| 2 | The Fellowship Of The Ring | male   | 971   | 3644   | 1995  |
| 3 | The Two Towers             | female | 331   | 0      | 401   |
| 4 | The Two Towers             | male   | 513   | 2463   | 3589  |
| 5 | The Return Of The King     | female | 183   | 2      | 268   |
| 6 | The Return Of The King     | male   | 510   | 2673   | 2459  |

## Your Turn 2

Use `pivot_wider()` to reorganize `table2` into four columns: `country`, `year`, `cases`, and `population`.

Create a new variable called `prevalence` that divides `cases` by `population` multiplied by 100000.

Pass the data frame to a `ggplot`. Make a scatter plot with `year` on the x axis and `prevalence` on the y axis. Set the color aesthetic (`aes()`) to `country`. Use `size = 2` for the points. Add a line geom.

```
1 table2
```

# Your Turn 2

```
1 table2 |>
2   pivot_wider(
3     names_from = type,
4     values_from = count
5   ) |>
6   mutate(prevalence = (cases / population) * 100000)
```

# A tibble: 6 × 5

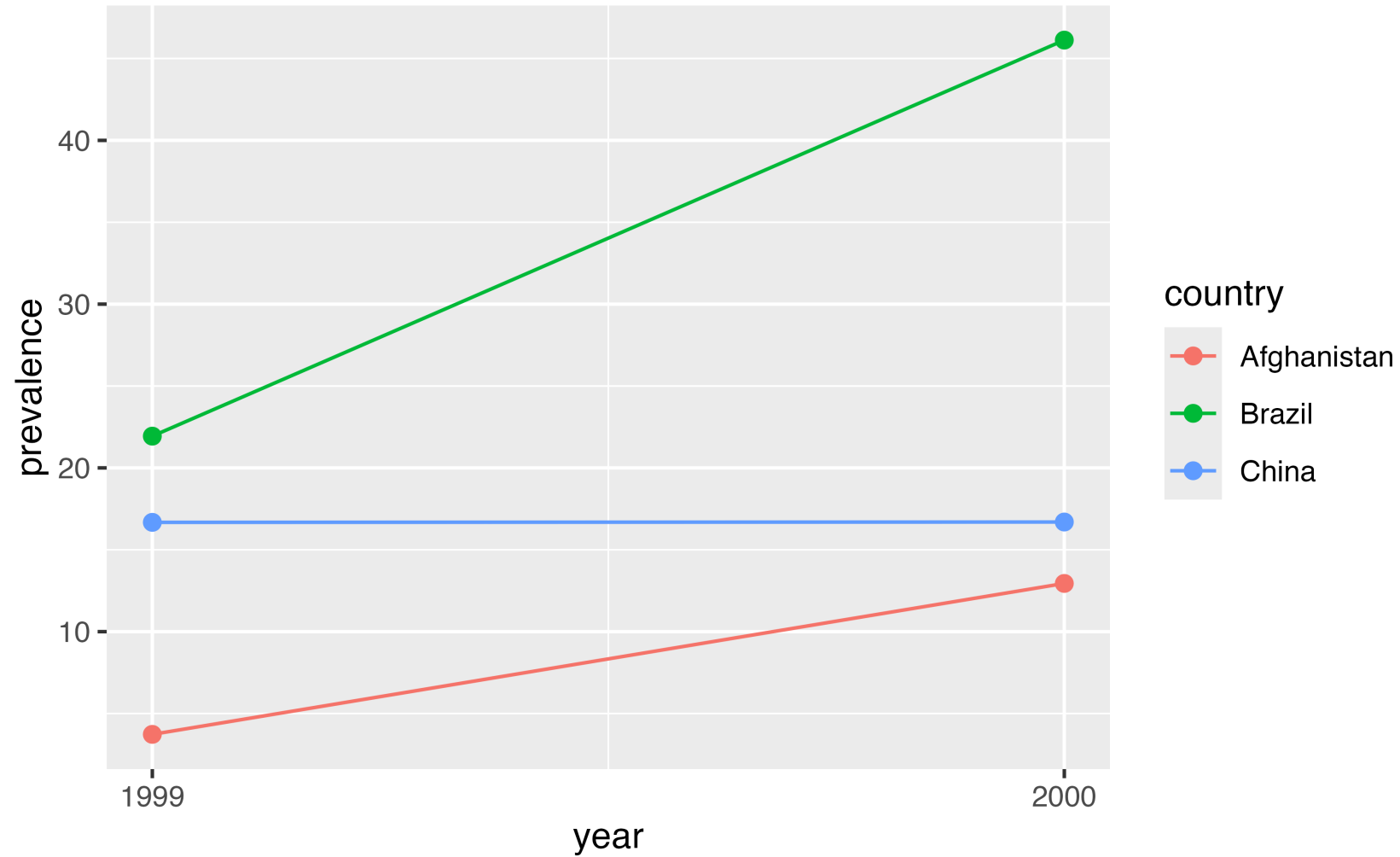
|   | country     | year  | cases  | population | prevalence |
|---|-------------|-------|--------|------------|------------|
|   | <chr>       | <dbl> | <dbl>  | <dbl>      | <dbl>      |
| 1 | Afghanistan | 1999  | 745    | 19987071   | 3.73       |
| 2 | Afghanistan | 2000  | 2666   | 20595360   | 12.9       |
| 3 | Brazil      | 1999  | 37737  | 172006362  | 21.9       |
| 4 | Brazil      | 2000  | 80488  | 174504898  | 46.1       |
| 5 | China       | 1999  | 212258 | 1272915272 | 16.7       |
| 6 | China       | 2000  | 213766 | 1280428583 | 16.7       |



# Your Turn 2

```
1 table2 |>
2   pivot_wider(
3     names_from = type,
4     values_from = count
5   ) |>
6   mutate(prevalence = (cases / population) * 100000) |>
7   ggplot(aes(x = year, y = prevalence, color = country)) +
8   geom_point(size = 2) +
9   geom_line() +
10  scale_x_continuous(breaks = c(1999L, 2000L))
```

# Your Turn 2



## Your Turn 3

**Pivot the 5th through 60th columns of `who` so that the names of the columns go into a new variable called `codes` and the values go into a new variable called `n`. Then select just the `country`, `year`, `codes` and `n` variables.**

```
1 who
```

# Your Turn 3

```
1 who |>
2   pivot_longer(
3     names_to = "codes",
4     values_to = "n",
5     cols = 5:60
6   ) |>
7   select(country, year, codes, n)
```

# Your Turn 3

```
# A tibble: 405,440 × 4
  country      year codes      n
  <chr>      <dbl> <chr>    <dbl>
1 Afghanistan 1980 new_sp_m014 NA
2 Afghanistan 1980 new_sp_m1524 NA
3 Afghanistan 1980 new_sp_m2534 NA
4 Afghanistan 1980 new_sp_m3544 NA
5 Afghanistan 1980 new_sp_m4554 NA
6 Afghanistan 1980 new_sp_m5564 NA
7 Afghanistan 1980 new_sp_m65 NA
8 Afghanistan 1980 new_sp_f014 NA
9 Afghanistan 1980 new_sp_f1524 NA
10 Afghanistan 1980 new_sp_f2534 NA
" # A tibble: 405,440 × 4"
```

# separate() / unite()

```
1 separate(  
2   <DATA>,  
3   <VARIABLE>,  
4   into = c("<VARIABLE1>", "<VARIABLE2>")  
5 )  
6  
7 unite(<DATA>, <VARIABLES>)
```

## Your Turn 4

Use the **cases** data below. Separate the **sex\_age** column into **sex** and **age** columns.

```
1 cases <- tribble(  
2   ~id,      ~sex_age,  
3   "1",      "male_56",  
4   "2",      "female_77",  
5   "3",      "female_49"  
6 )  
7 separate(_____, _____, into = c("_____", "_____"))
```

# Your Turn 4

```
1 cases <- tribble(  
2   ~id,      ~sex_age,  
3   "1",      "male_56",  
4   "2",      "female_77",  
5   "3",      "female_49"  
6 )  
7 separate(cases, sex_age, into = c("sex", "age"))
```

```
# A tibble: 3 × 3  
  id      sex      age  
<chr> <chr> <chr>  
1 1      male    56  
2 2      female  77  
3 3      female  49
```



# Your Turn 4

```
1 cases <- tribble(  
2   ~id,      ~sex_age,  
3   "1",      "male_56",  
4   "2",      "female_77",  
5   "3",      "female_49"  
6 )  
7 separate(  
8   cases,  
9   sex_age,  
10  into = c("sex", "age"),  
11  convert = TRUE  
12 )
```

```
# A tibble: 3 × 3  
  id    sex    age  
  <chr> <chr>  <int>  
1 1    male    56  
2 2  female    77  
3 3  female    49
```

## ***Your Turn 5: Challenge!***

There are two CSV files in this folder containing SEER data in breast cancer incidence in white and black women. For both sets of data:

**Import the data**

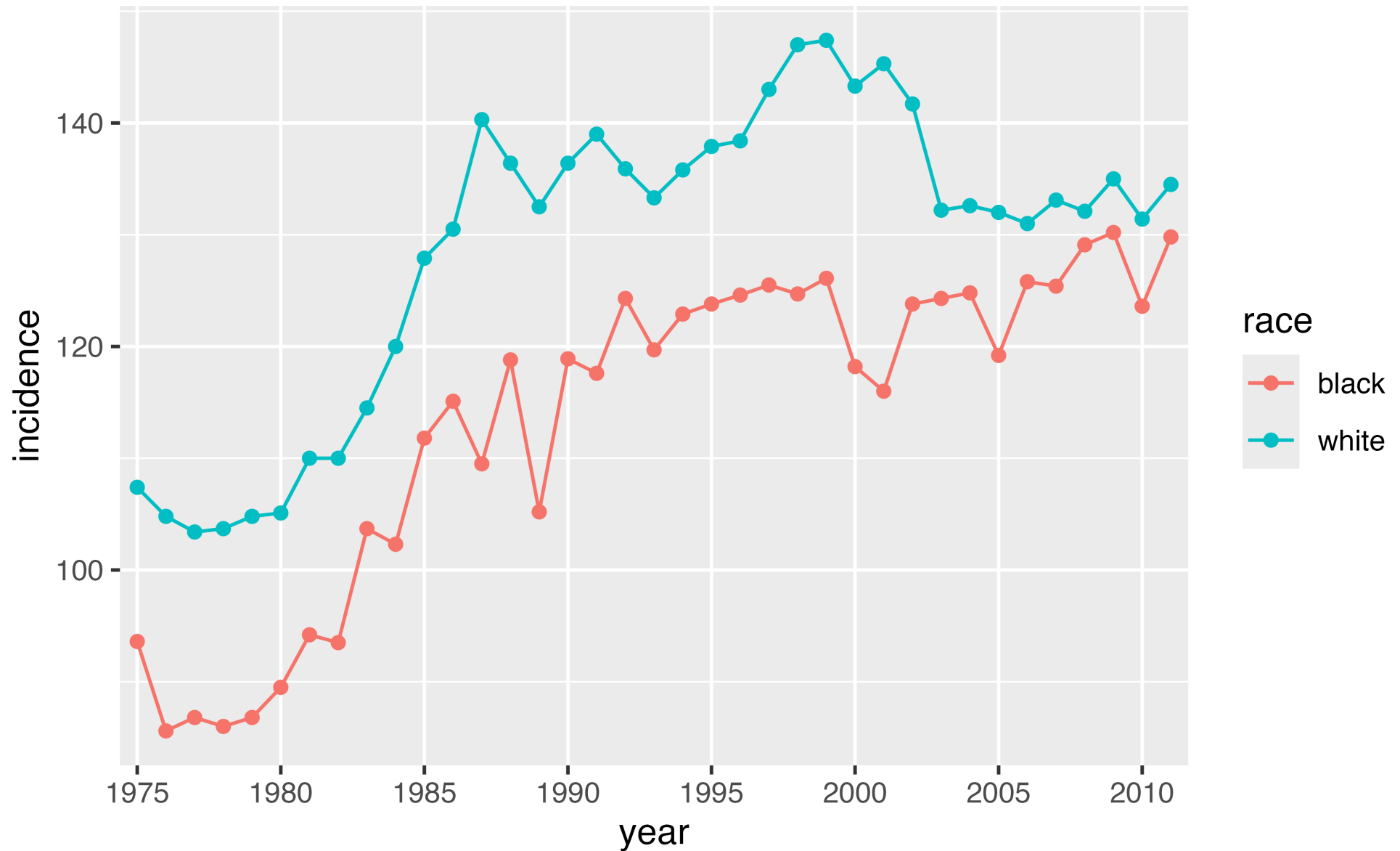
**Pivot the columns into 2 new columns called `year` and `incidence`**

**Add a new variable called `race`. Remember that each data set corresponds to a single race.**

**Bind the data sets together using `bind_rows()` from the dplyr package. Either save it as a new object or pipe the result directly into the ggplot2 code.**

**Plot the data using the code below. Fill in the blanks to have `year` on the x-axis, `incidence` on the y-axis, and `race` as the color aesthetic.**

# ***Your Turn 5: No solution*** 🍆



# Other neat tidyr tools

## Uncounting frequency tables

```
1 lotr |>
2   pivot_longer(
3     names_to = "sex",
4     values_to = "count",
5     cols = c(female, male)
6   ) |>
7   uncount(count)
```

# Other neat tidyr tools

```
# A tibble: 21,245 × 3
```

```
  film                                race  sex  
  <chr>                                <chr> <chr>  
1 The Fellowship Of The Ring Elf      female  
2 The Fellowship Of The Ring Elf      female  
3 The Fellowship Of The Ring Elf      female  
4 The Fellowship Of The Ring Elf      female  
5 The Fellowship Of The Ring Elf      female  
6 The Fellowship Of The Ring Elf      female  
7 The Fellowship Of The Ring Elf      female  
8 The Fellowship Of The Ring Elf      female  
9 The Fellowship Of The Ring Elf      female  
10 The Fellowship Of The Ring Elf      female
```

```
" 1 24 225 more rows"
```

# Other neat tidyr tools

## *Work with data frames*

`crossing()` and `expand()`

`nest()` and `unnest()`

# Other neat tidyr tools

## *Work with missing data*

`complete()`

`drop_na()` and `replace_na()`

# Resources

**R for Data Science:** A comprehensive but friendly introduction to the tidyverse. Free online.

**Posit Recipes:** Common code patterns in R (with some comparisons to SAS)